

Cloud Computing: Unit 1 Principles and Virtualization

VTU 2022 Scheme CBCS - BCS601 Module 1

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. NIST Definition of Cloud Computing: 5 Essential Characteristics
2. Cloud Service Models: Software as a Service [SaaS], PaaS, IaaS
3. Cloud Deployment Models: Public, Private, Hybrid, and Community Cloud
4. Virtualization Reference Model & Hardware-Assisted Virtualization
5. Hypervisors: Type-1 [Bare-Metal] vs Type-2 [Hosted] Hypervisor architectures
6. CPU, Memory, and I/O Virtualization Mechanisms

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Explain NIST cloud architecture with neat block diagram [10 Marks]
- Q2. Differentiate Type 1 and Type 2 hypervisors with suitable examples [10 Marks]
- Q3. Write short notes on Virtual Machine lifecycle and migration [10 Marks]